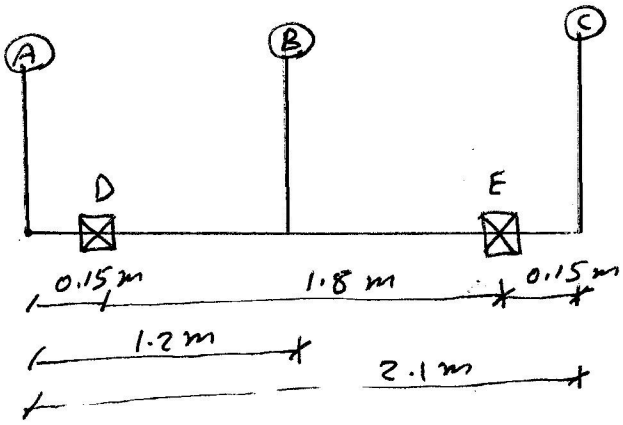
2. A shaft carries three pulleys, A, B and C, the pulleys B and C being 1·2 m and 2·1 m from A. The pulleys are out-of-balance to the extent of 0·03, 0·05 and 0·04 kg m respectively and are keyed to the shaft so as to give static balance.

Find the relative angular positions of the radii which define the unbalance. If the shaft is supported in bearings 1·8 m apart, find the dynamic load on each bearing when the shaft makes 300 rev/min.

Plane No.	mass kg	r (m)	C.f ÷ ω^2 kg·m	l (m)	Couple ÷ ω^2 kg·m²	θ
A	m_A	r_A	0.03	−0.15	−0.045	0
B	m_B	r_B	0.05	1.05	0.0525	
C	m_C	r_C	0.04	1.95	0.078	
R.P D	—	—	R_D	0	0	
E	—	—	R_E	1.8	1.8 R_E	



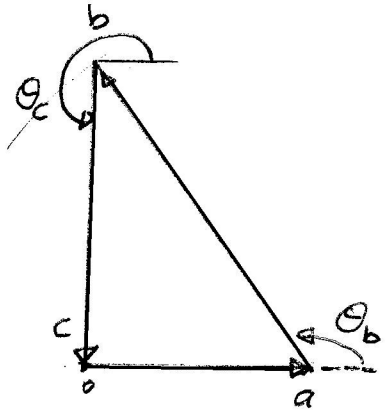
For Static Balance ⇒ $R_D = R_E = 0$

C.f Diagram

Scale ⇒ 0.01 kg·m = 1 cm

∴ $\theta_b = 127°$ Relative to A

$\theta_c = 270$ " "



Couple Diagram

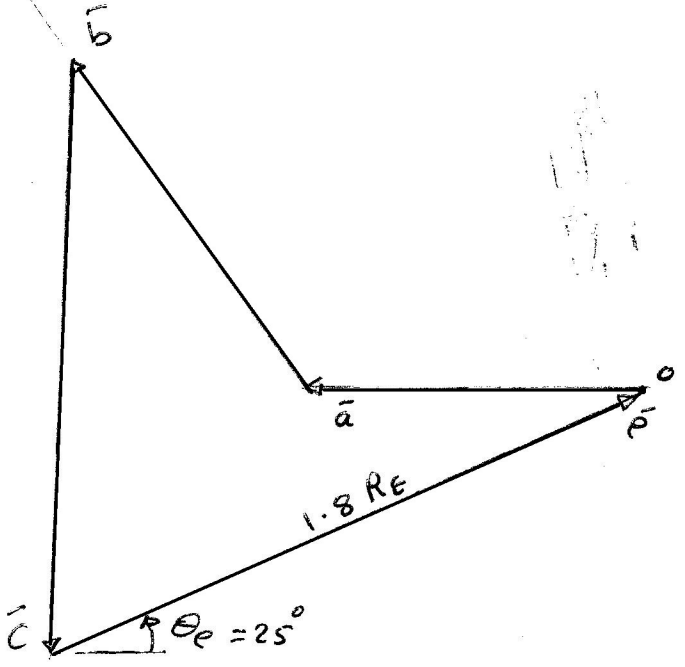
Scale ⇒ 0.01 kg·m² = 1 cm

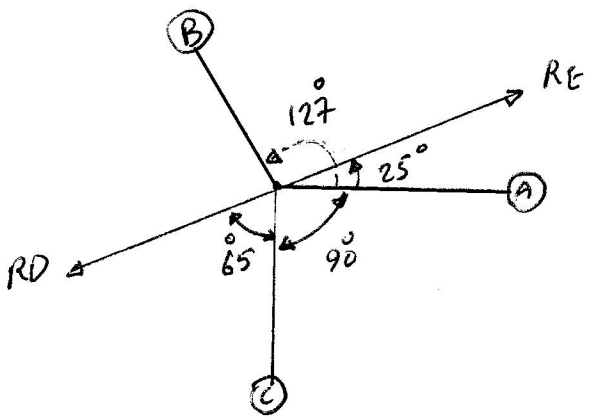
∴ $1.8 R_E = 8.5 (0.01)$

$R_E = 0.04723$ kg·m

$R_E = 0.04723 \left(300 \times \frac{2\pi}{60}\right)^2 = 46.6 N$

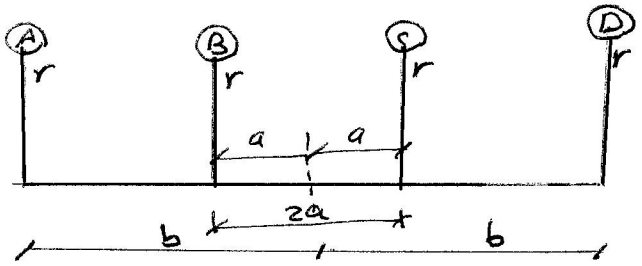
$R_E = R_D = 46.6 N$



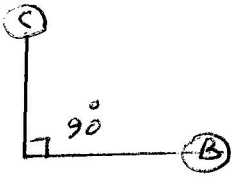


3. Four masses, at equal radii and rotating in parallel planes, are attached to a shaft. In each of the end planes, spaced a distance of $2b$ apart, there is a mass of 12 kg. The inner planes, spaced a distance $2a$ apart, are symmetrical with the end planes and each contains a mass of 15 kg. If the masses on the inner planes are at right angles to each other, find the ratio of a to b and the relative angular positions of the other masses for complete balance.

Plane No.	mass kg	r (m)	c.f $\div \omega^2$ kg.m	ℓ (m)	couple $\div \omega^2$ mrℓ	θ
R.P A	12	r	12r	0	0	θ_A
B	15	r	15r	(b-a)	15r(b-a)	0
C	15	r	15r	(b+a)	15r(b+a)	90°
D	12	r	12r	2b	24rb	θ_D



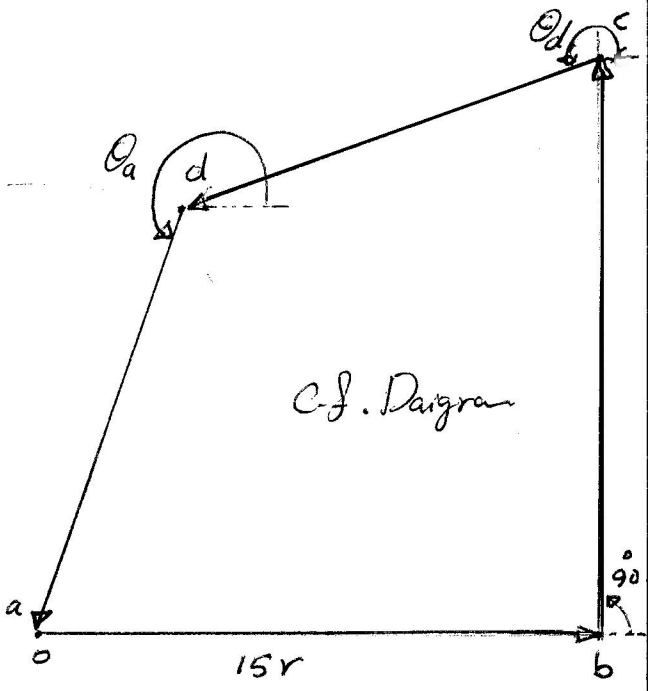
C.f Diagram



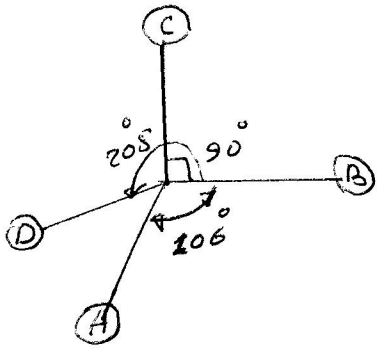
Scale ⇒ 2r = 1cm

∴ $\theta_d = 208°$ from B

$\theta_a = 254°$ from B



Couple Diagram



بدوم مقیاس رسم فان c ⊥ B وبالتالي يتكون مثلث قائم الزاوية :.

$$\therefore (24rb)^2 = [15r(b+a)]^2 + [15r(b-a)]^2$$

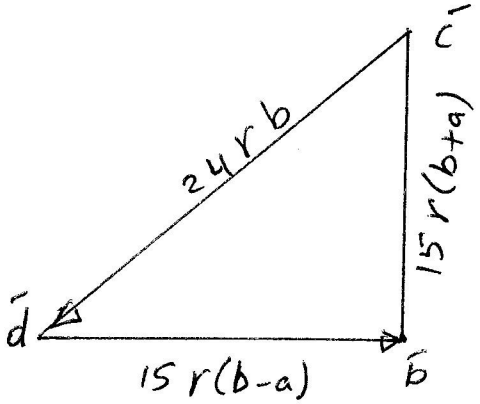
$$\frac{24^2 r^2}{15^2 r^2} b^2 = (b+a)^2 + (b-a)^2$$

$$\therefore 2.56\, b^2 = b^2 + 2ba + a^2 + b^2 - 2ab + a^2$$

$$\therefore 0.56 b^2 = 2a^2$$

$$\therefore \frac{a}{b} \simeq 0.53$$

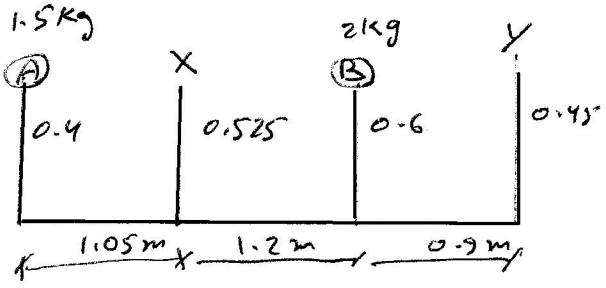
Ans.



4. A turbine rotor is found to be out of balance to the extent of 1·5 kg at 0·45 m radius in the plane AA and 2 kg at 0·6 m radius in the plane BB, the relative angular positions being given in the end view, Fig. 7.25. It is desired to balance these masses by a mass in each of the planes XX and YY at radii of 0·525 m and 0·45 m respectively.

Determine the magnitude and positions of these masses and show their positions in an end view.

Plane No.	mass kg	r(m)	C.f ÷ ω^2 kg·m	ℓ (m)	couple ÷ ω^2 kg·m²
A	1.5	0.4	0.6	−1.05	−0.63
B	2	0.6	1.2	1.2	1.44
R-PX	M_x	0.525	$0.525 M_x$	0	0
Y	M_y	0.45	$0.45 M_y$	2.1	$0.945 M_y$

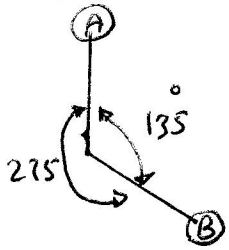


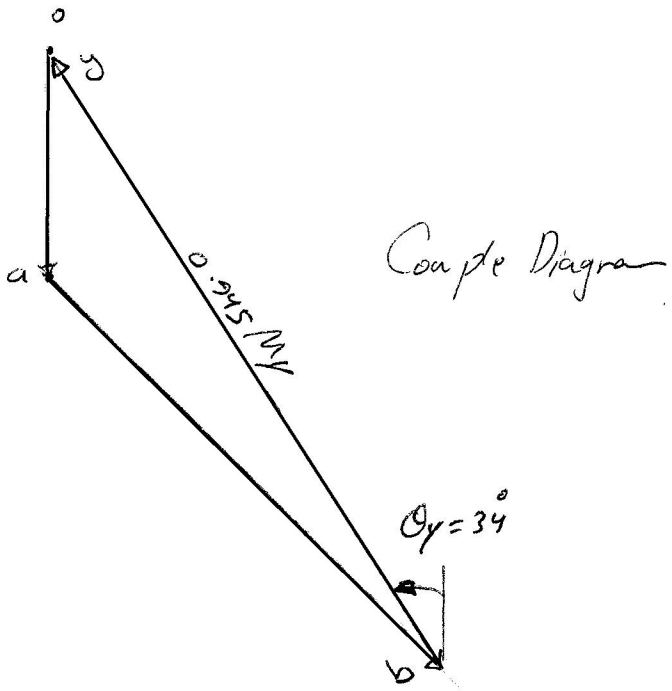
Couple Diagram

Scale ⇒ 0.2 kg·m² = 1cm

$\therefore 0.945 M_y = 9.8(0.2)$

$\therefore M_y = 2.074$ kg





Couple Diagram

C.f. Diagram

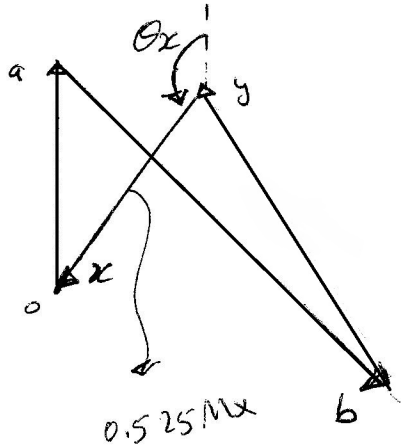
Scale ⇒ 0.2 kg·m = 1cm

$\therefore F_y = 0.45 \omega^2 M_y = 0.45(2.074) = 0.934$ kg·m

$\therefore 0.525 M_x = 3.3(0.2)$

$\therefore M_x = 1.26$ kg

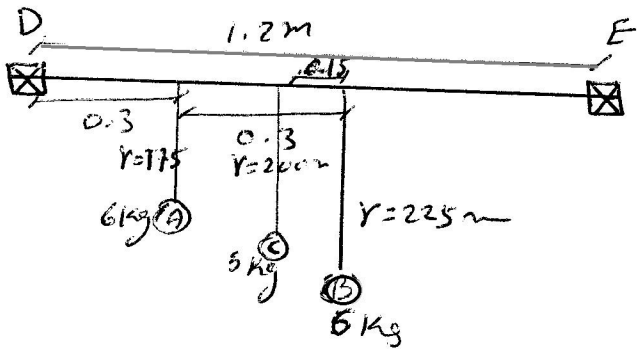
Ans.



C.f. Diagram

6. A horizontal shaft supported in bearings 1·2 m apart carries two masses, A and B, each of mass 6 kg, placed at distances of 0·3 m and 0·6 m respectively from one of the bearings. Static balance is obtained by the addition of a balance mass of 5 kg acting at a radius of 200 mm, the position of the balance mass being midway between A and B. If the radii of the centres of mass of A and B are 175 mm and 225 mm respectively, find the relative angular positions of the three masses, and also the magnitude of the unbalanced couple acting on the bearings when the shaft rotates at 100 rev/min.

plane No.	mass kg	r (m)	cf ÷ ω^2 kg.m	l (m)	couple ÷ ω^2 kg.m^2
A	6	0.175	1.05	0.3	0.315
B	6	0.225	1.35	0.6	0.81
C	5	0.2	1	0.45	0.45
R.P D	—	—	RD	0	0
E	—	—	RE	1.2	1.2 RE



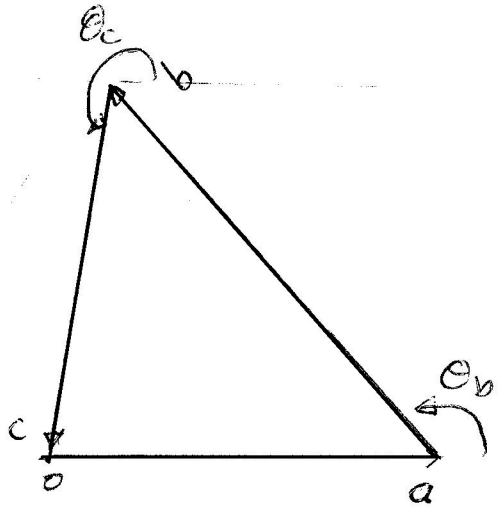
For Static Balance $\Rightarrow R_E = R_D = 0$

C.F. Diagram

Scale $\Rightarrow$ 0.2 kg.m = 1cm

$\theta_b = 131^\circ$ from A

$\theta_c = 259$ from A



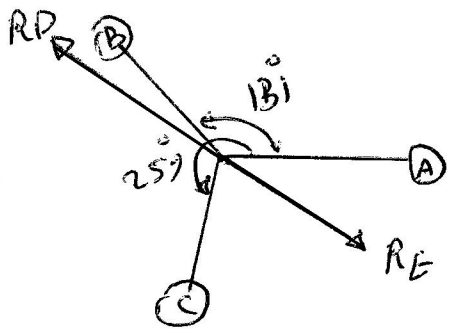
Couple Diagram

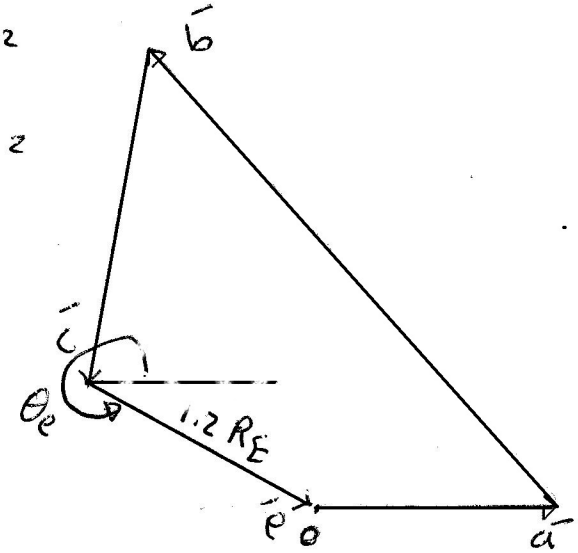
Scale $\Rightarrow$ 0.1 kg.m^2 = 1cm

$1.2 R_E$ = Couple = 3.3 (0.1) kg m^2

$\therefore$

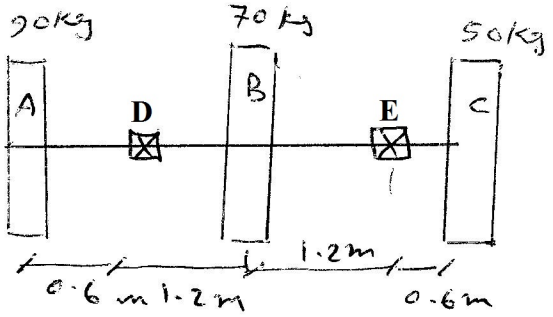
$$\text{Couple} = 3.3(0.1)\left(\frac{100 \times 2\pi}{60}\right)^2$$





7. A shaft is supported in two bearings 2·4 m apart and projects 0·6 m beyond the bearings at each end. The shaft carries three pulleys, one at each end and one at the middle of its length. The end pulleys have masses of 90 kg and 50 kg and their centres of mass are at 3·75 mm and 5 mm respectively from the shaft axis. The centre pulley has a mass of 70 kg and its centre of mass is 6·25 mm from the shaft axis. If the pulleys are arranged so as to give static balance, determine the dynamic forces produced on the bearings when the shaft rotates at 300 rev/min.

plane NO.	mass kg	r(m)	cf ÷ ω^2 kg·m	l(m)	couple ÷ ω^2 kg·m²
A	90	0·00375	0.3375	−0·6	−0·2025
B	70	0·00625	0.4375	1.2	0·525
C	50	0·005	0·25	3	0·75
RP D	—	—	R_D	0	0
E	—	—	R_E	2·4	2·4 R_E



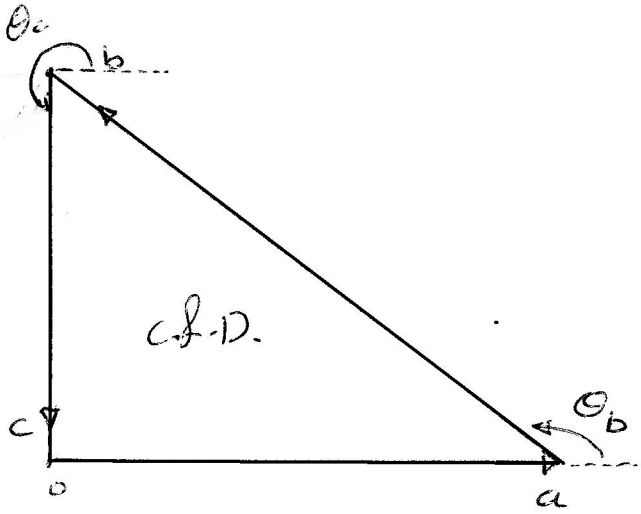
for Static Balance

$\therefore R_D = R_E = 0$

Scale ⇒ 0·05 kg·m = 1cm

$\therefore \theta_b = 144°$ from A

$\theta_c = 270$ from A



Couple Diagram

Scale ⇒ 0·1 kg·m² = 1cm

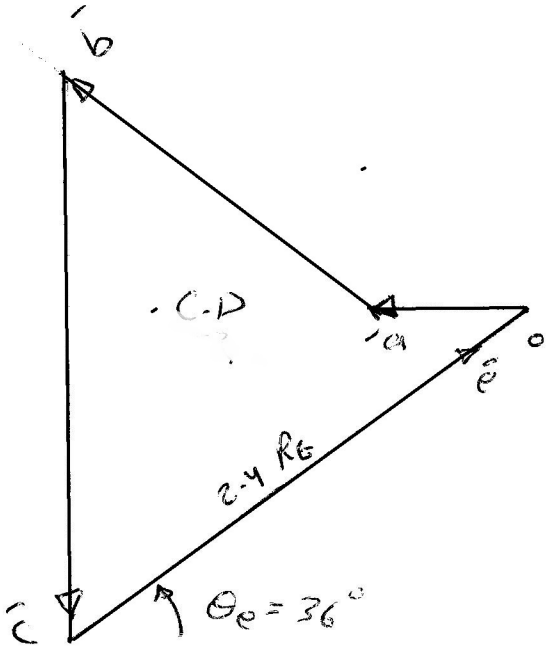
$\therefore 2.4 R_E = 7.6 * 0.1$

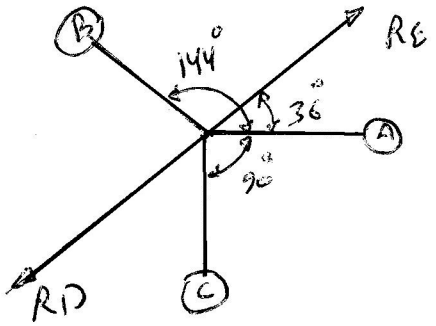
$\therefore R_E = 0.3167$ kg·m

$\therefore R_E = 0.3167 \times \left(\frac{300 \times 2\pi}{60}\right)^2$

$= 312.6$ N

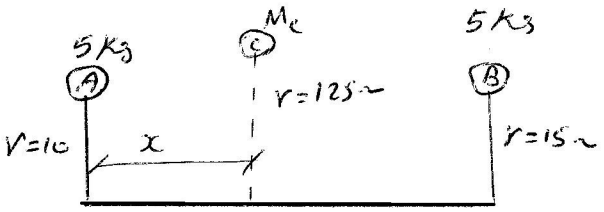
$\therefore R_E = R_D = 312.6$ N





9. The out-of-balance of a machine rotor is equivalent to 5 kg at 10 mm radius in one plane A, together with an equal mass at 15 mm radius in a second plane B. AB = 375 mm and the two radii are at 120°. Find the mass required in a third plane C at a radius of 125 mm and its angular position with respect to the given radii so that there is no resultant out-of-balance force. Find also the position of C along the axis for the residual couple to be a minimum and the value of this couple when the speed is 500 rev/min.

Plane No.	mass Kg	r(m)	C.f $\div\omega^2$ Kg·m	ℓ (m)	couple $\div\omega^2$ Kg·m²	θ
A	5	0.01	0.05	0	0	0
B	5	0.015	0.075	0.375	0.0281	120°
C	M_c	0.125	$0.125M_c$	x	$0.125M_cx$	θ_c



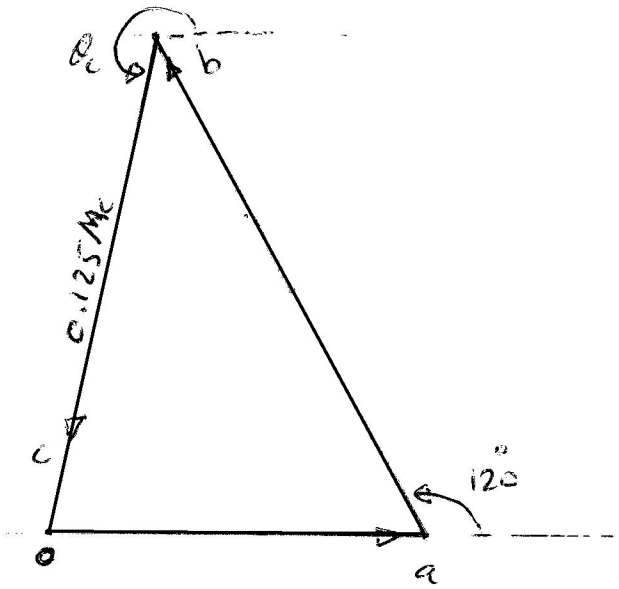
C.f. Diagram

Scale ⇒ 0.01 kg·m = 1cm

$\therefore \theta_c = 255°$ from A

$0.125 M_c = 6.6(0.01)$

$\therefore M_c = 0.528$ kg



Couple Diagram

Scale ⇒ 0.005 kg·m² = 1cm

$4.2(0.005) = 0.125 M_c x$

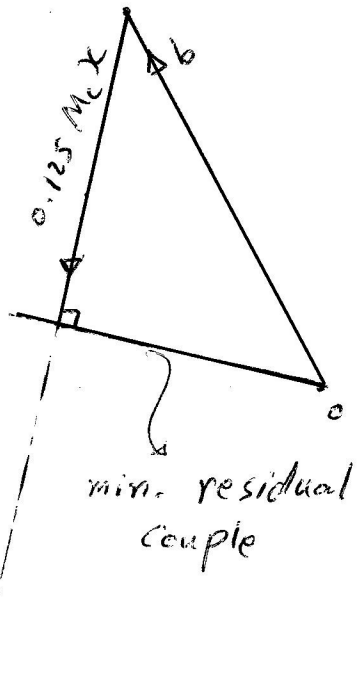
$M_c = 0.528$ kg

$\therefore x = 0.318\ m = 318\ mm$

Min. residual couple $= 3.7(0.005) \times \left(500 \times \frac{2\pi}{60}\right)^2$

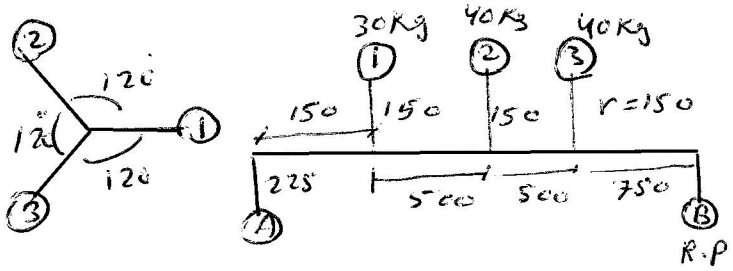
$= 50.6$ N·m

Ans.



10. A three-throw crankshaft has double-webbed cranks of 150 mm radius set at 120° to each other and equally spaced with a pitch of 500 mm. The rotating masses at crank radius are: No. 1, 30 kg; No. 2, 40 kg; No. 3, 40 kg. Balance is to be effected by a balance mass attached to the outside web of crank No. 1, with a centre of mass 150 mm from the central plane of the crank and at a radius of 225 mm from the centre of the shaft; and also by removing material, at a radius of 750 mm, from a wheel fixed 750 mm beyond the central plane of crank No. 3.

Determine the masses to be fitted and removed and their angular positions relative to crank No. 1.



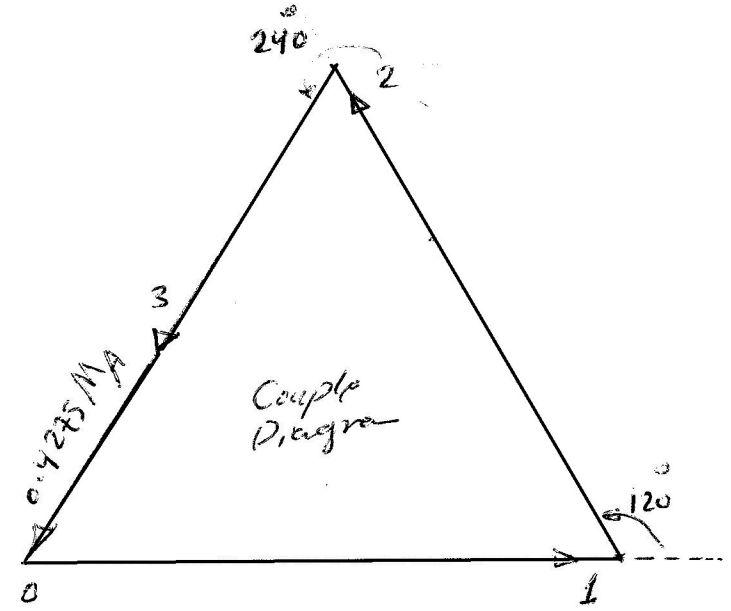
plane No.	mass Kg	r(m)	cf ÷ ω² Kg.m	ℓ (m)	couple Kg.m²
1	30	0.15	4.5	1.75	7.875
2	40	0.15	6	1.25	7.5
3	40	0.15	6	0.75	4.5
A	$+m_A$	0.225	$0.225M_A$	1.9	$0.4275M_A$
R.P B	$-m_B$	0.75	$-0.75M_B$	0	0

Couple Diagram

Scale ⇒ 1 Kg.m² = 1cm

$\therefore 0.4275M_A = 3.3(1)$

$\therefore M_A = 7.7$ Kg



c.f. Diagram

Scale ⇒ 1 Kg.m = 1cm

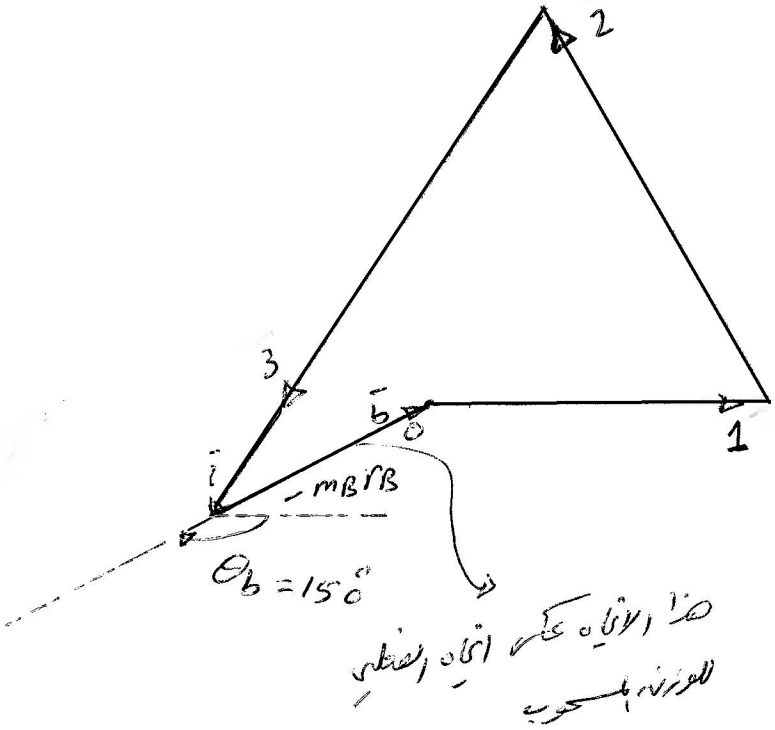
$0.225M_A = 1.7325$ Kg.m

$\therefore m_B(0.75) = 3(1)$

$\therefore m_B = 4$ Kg

Ans.

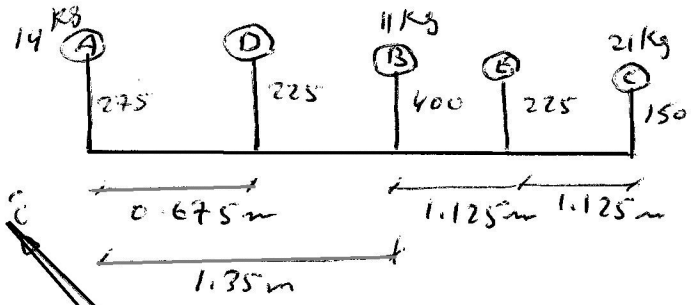
$\theta_b = 150°$



11. Three rotating masses, A = 14 kg, B = 11 kg and C = 21 kg, are carried on a shaft, with centres of mass 275 mm, 400 mm and 150 mm respectively from the shaft axis. The angular positions of B and C are 60° and 135° respectively from A, measured in the same direction. The distance between the planes of rotation of A and B is 1·35 m, and between those of A and C is 3·6 m, B and C being on the same side of A.

Two balance masses are to be fitted, each with its centre of mass 225 mm from the shaft axis, in planes midway between those of A and B and of B and C. Determine the magnitude and angular position with respect to A of each balance mass.

Plane No.	Mass kg	r (m)	c.f. ÷ ω^2 kg·m	ℓ (m)	couple ÷ ω^2 kg·m²	θ
A	14	0.275	3.85	−0.675	−2.6	0
B	11	0.4	4.4	0.675	2.97	60°
C	21	0.15	3.15	2.925	9.214	135°
R.P. D	M_D	0.225	$0.225M_D$	0	0	θ_D
E	M_E	0.225	$0.225M_E$	1.8	$0.405M_E$	θ_E

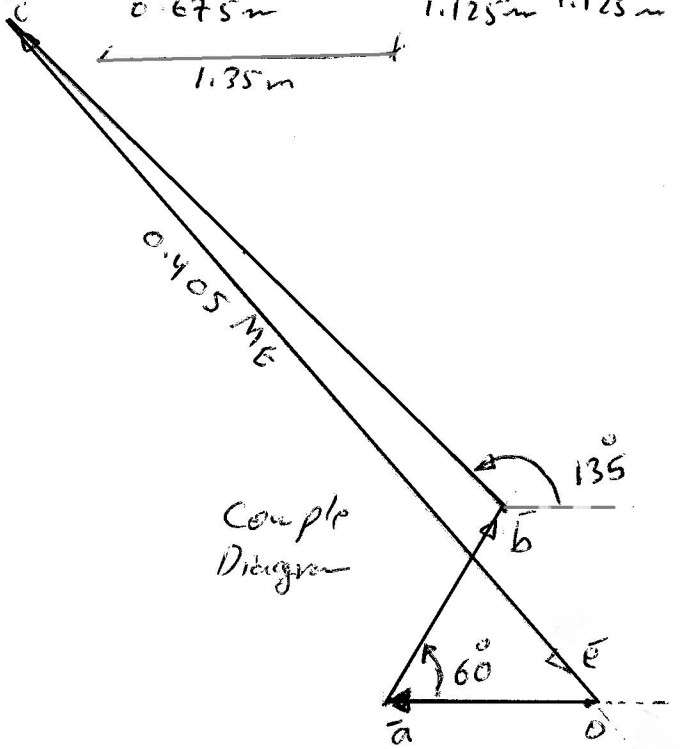


Couple Diagram

Scale ⇒ 1 kg·m² = 1 cm

$\therefore 11.8(1) = 0.405 M_E$

$M_E = 29.14$ kg



Couple Diagram

c.f. Diagram

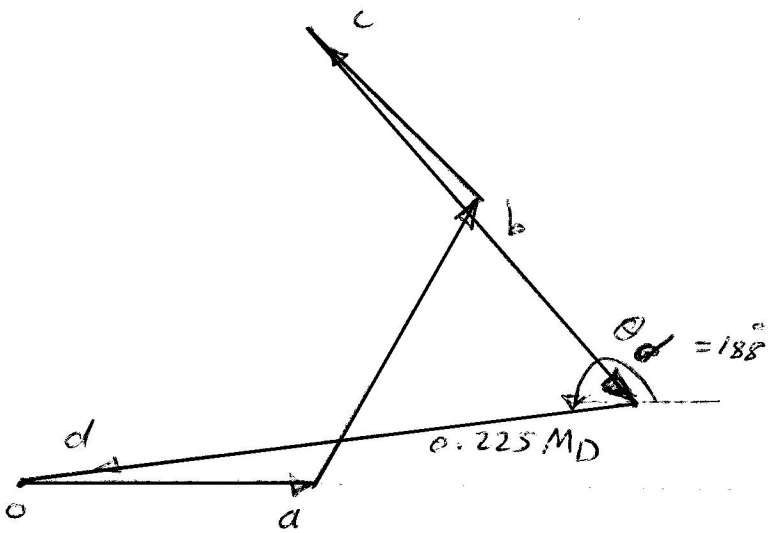
Scale ⇒ 1 kg·m = 1 cm

$0.225 M_E = 6.557$ kg·m

$\therefore 3.1(1) = 0.225 M_D$

$\therefore M_D = 36$ kg

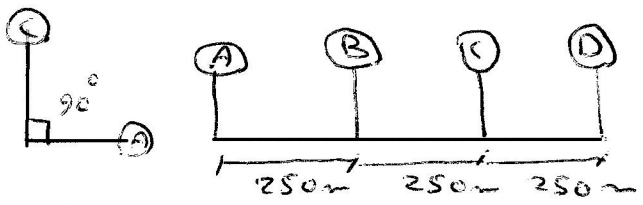
$\theta_d = 188°$



13. A shaft carries four wheels, A, B, C and D, equally pitched 250 mm apart. The unbalance (mr) values for A and C are, respectively, 0·05 kg m and 0·06 kg m, and the line of unbalance in C is at 90° to that of A, which may be taken as the reference direction. The out-of-balance amounts for B and D are initially unknown, but the complete rotor is dynamically balanced by adding a mass of 0·324 kg to wheel B at a radius of 0·6 m and at an angle of 215° to the reference, and by removing material of mass 0·08 kg from D at a radius of 0·45 m and at an angle of 120°.

Determine the initial and final unbalance values for B and D.

Plane No.	mass kg	r (m)	C.f. $\div \omega^2$ kg·m	ℓ (m)	couple $\div \omega^2$ kg·m²	θ
A	m_A	r_A	0.05	0.75	0.0375	0
B	m_B	r_B	$m_B r_B$	0.5	$0.5\, m_B r_B$	θ_B
	+0.324	0.6	+0.1944	0.5	0.0972	215°
C	m_C	r_C	0.06	0.25	0.015	90°
R.P. = D	m_D	r_D	$m_D r_D$	0	0	θ_D
	−0.08	0.45	−0.036	0	0	120°



Couple Diagram

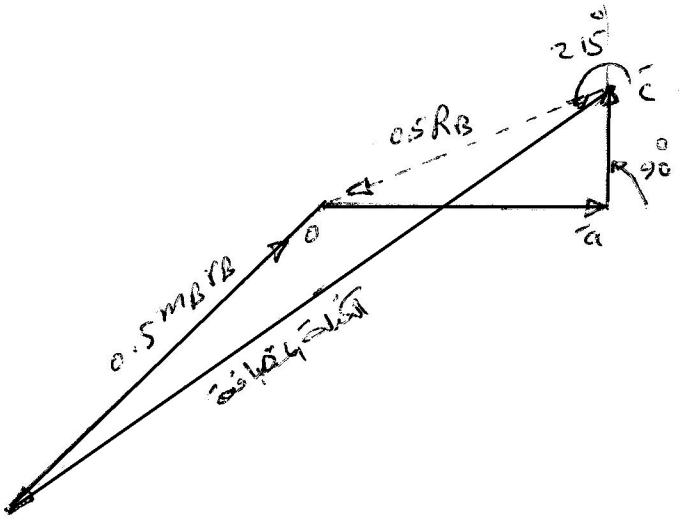
Scale 0.01 kg·m² = 1 cm

$\therefore 5.7(0.01) = 0.5\, m_B r_B$

$\therefore m_B r_B = 0.114$ kg·m (Initial)

$0.5\, R_B = 4(0.01) = 0.04$ kg·m²

$\therefore R_B = \dfrac{0.04}{0.5} = 0.08$ kg·m (final)



C.f. Diagram

Scale ⇒ 0.01 kg·m = 1 cm

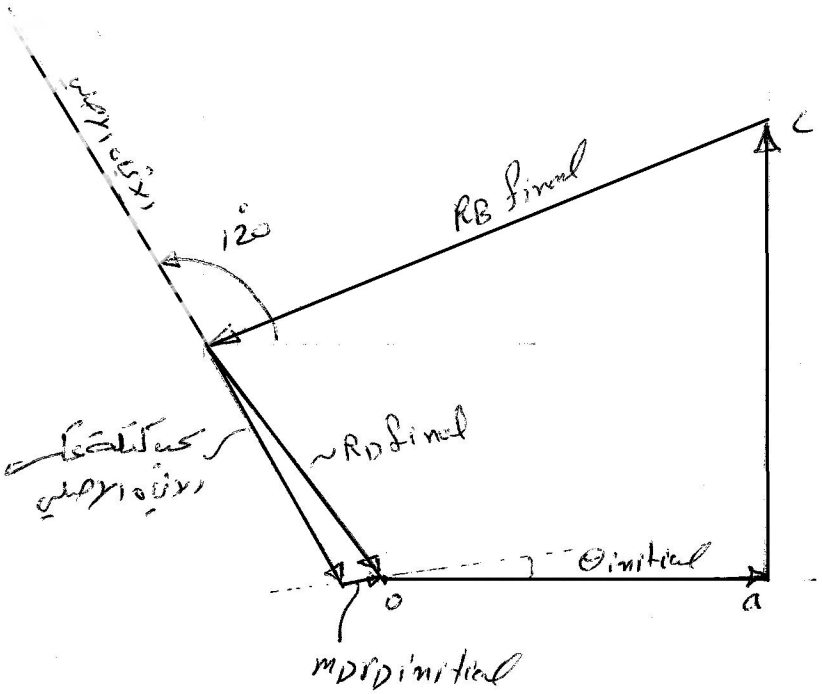
$m_B r_B$ final = 0.08 kg·m

$0.6(0.01) = m_B r_B$ Initial

$m_D r_D$ Initial = 0.006 kg·m

$m_B r_B$ final = R_D

$\therefore R_D = 3.8(0.01) = 0.038$ kg·m



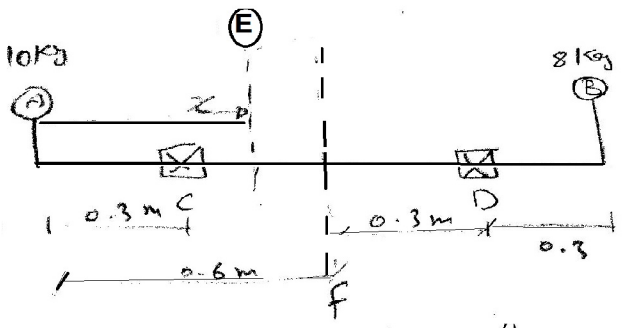
14. A shaft turning at a uniform speed carries two uniform discs A and B of masses 10 kg and 8 kg respectively. The centres of mass of the discs are each 2·5 mm from the axis of rotation. The radii to the centres of mass are at right angles. The shaft is carried in bearings C and D between A and B such that AC = 0·3 m, AD = 0·9 m and AB = 1·2 m.

It is required to make the dynamic loading on the bearings equal and a minimum for any given shaft speed by adding a mass at a radius of 25 mm in a plane E.

Determine: (a) the magnitude of the mass in plane E and its angular position relative to the radius through the centre of mass in plane A, (b) the distance of plane E from plane A, and (c) the dynamic loading on each bearing when the mass in plane E has been attached and the shaft turns at 200 rev/min. (U. Lond.)

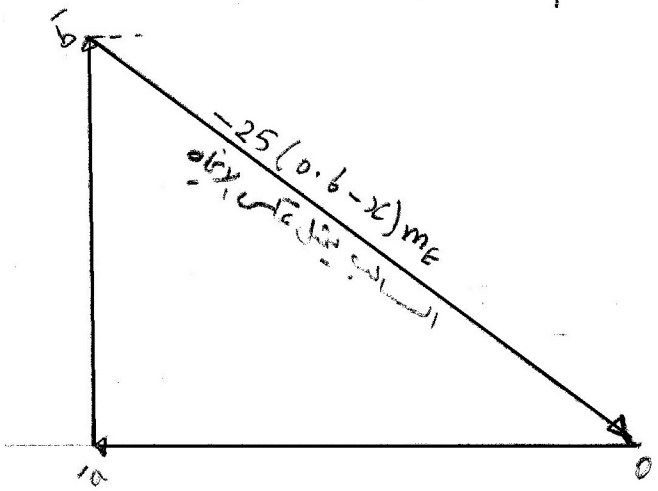
$N = 200$ r.p.m , $f_c = f_D$ and minimum

If Bearing loads in same direction



assume a new plane f

Plane No.	mass kg	r mm	$mr \div w^2$ kg·mm	h m	$mrL \div w^2$ kg·mm·m	Θ
A	10	2·5	25	−0·6	−15	0
B	8	2·5	20	0·6	12	90°
C	—	—	f_c	−0·3	$-0·3 f_c$	Θ_c
D	—	—	f_D	0·3	$0·3 f_D$	Θ_D
E	m_E	25	$25 m_E$	$-(0·6-x)$	$-25(0·6-x)m_E$	Θ_E
R.P.f						



Couple Diagram

Scale ⇒ 1 kg·mm·m = 2 cm

$\therefore 25(0·6-x)m_E = 9.4 \times 2$

$\therefore (0·6-x)m_E = 0.752$ — (1)

C.f. D

Scale ⇒ 1 kg·mm = 4 cm

$\therefore 25 m_E = 1.9 \times 4 \Rightarrow m_E = 0.3$ kg

$\therefore (0·6-x)0.3 = 0.752$

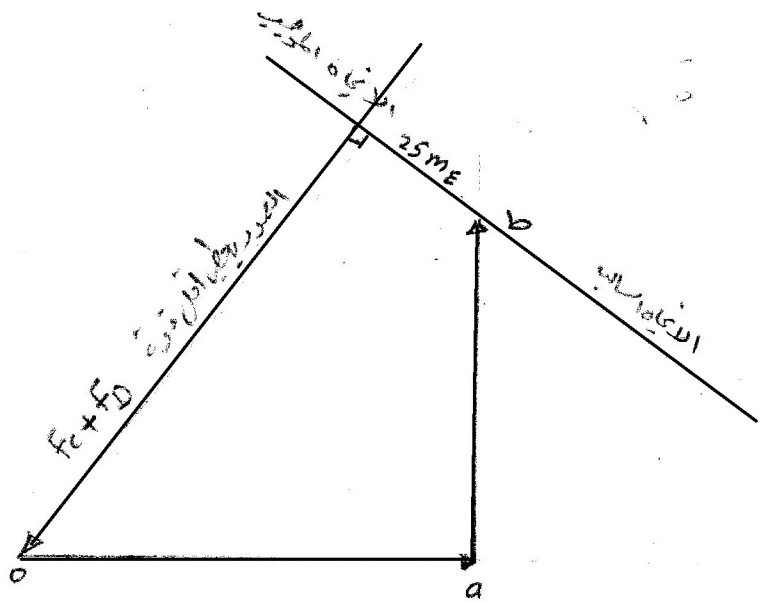
$\therefore x = 1.91$ m

$f_c = f_D$

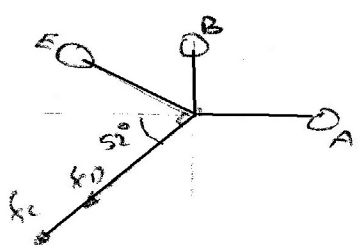
$f_c + f_D = 7.9 \times 4$

$\therefore 2 f_c = 31.6 \Rightarrow$

$f_c = f_D = 15.8$ kg·mm



$\therefore f_c = f_D = 15.8 \times 10^{-3} \times \left(\frac{2\pi}{60} \times 200\right)^2 = 6.93$ N

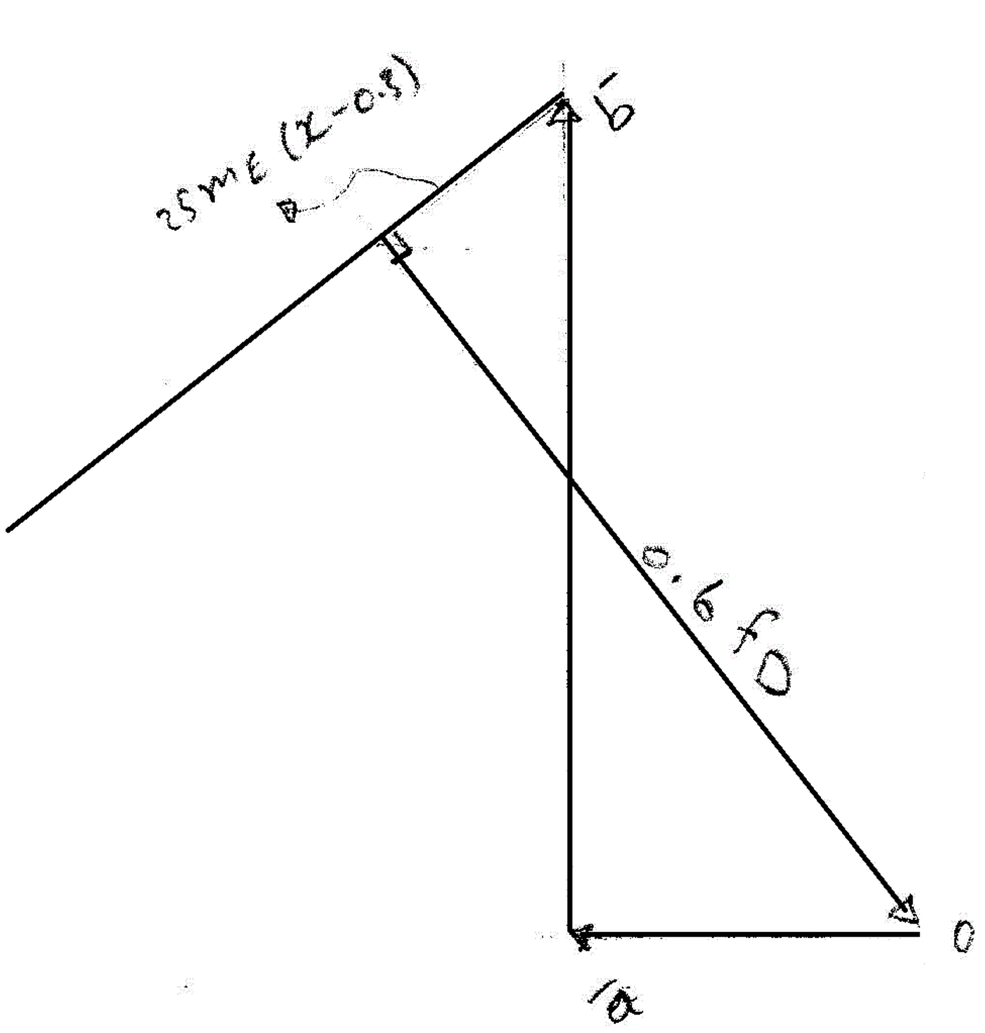


تكملة سؤال 14

$f_C = f_D$

for bearing loads in opposite direction

plane	mass Kg	r mm	$mr \div \omega^2$ kg.mm	h m	$mrL \div \omega^2$ kg.mm.m	θ
A	10	2.5	25	−0.3	−7.5	0
B	8	2.5	20	+0.9	18	90°
R.P C	—	—	f_C	0	0	θ_C
D	—	—	f_D	0.6	$0.6 f_D$	θ_D
E	m_E	25	$25 m_E$	$x-0.3$	$25 m_E (x-0.3)$	θ_E



C.f.D

Scale ⇒ 1 kg.mm = 4 cm

$\therefore 25 m_E = 8 * 4 \Rightarrow m_E = 1.28$ kg

Couple Diagram

Scale ⇒ 1 kg.mm.m = 2.5 cm

$\therefore 25 m_E (x-0.3) = 2.1 * 2.5$

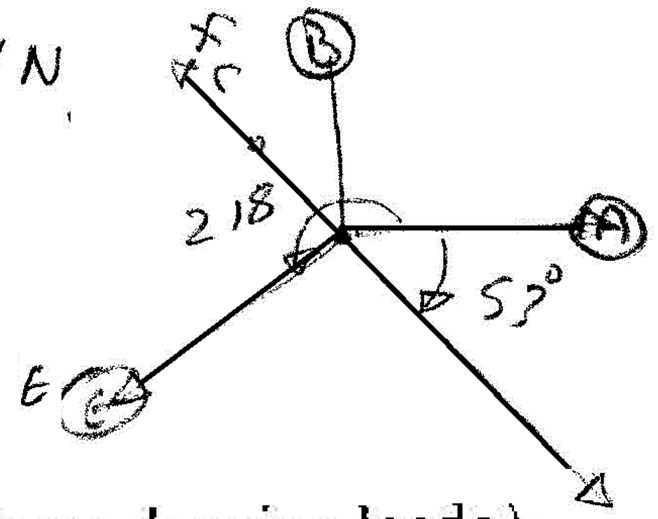
$\therefore 25(1.28)(x-0.3) = 2.1 * 2.5$

$\therefore x = 0.464$ m

$\therefore 7.7 + 2.5 = 0.6 f_D \Rightarrow$

$f_D = 32.1$ kg.mm

$\therefore f_D = f_C = 32.1 * 10^{-3} * \left(\frac{2\pi}{60} \times 200\right)^2 = 14$ N



(The first set of answers gives a lower value for the minimum bearing loads.)